Find the type of Supplied char.

```
char ch;
puts("Enter a char:");
scanf("%c",&ch); // 2'
isdigit(ch) ? pf("D") : 0;
isupper(ch) ? pf("U") : 0;
islower(ch) ? pf("L") : 0;
```

stdin → char → UPPER / lower / numeric / Spl

R1: 0–47 (48–57 D)
R2: 58–64
R3: 91–96 (65–90 U)
R4: 123–127 (97–122 L)

```
(R1 || R2 || R3 || R4) ? pf("Spl") : 0;
```

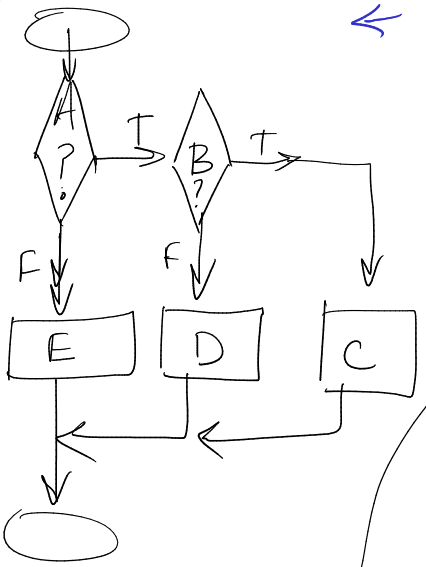
```
!(D || U || L) ? pf("-") : 0;
```

ch = ↵ => 10

```
!(isdigit(ch) || isupper(ch) || islower(ch)) ? pf("Spl") : 0;
```

0 0 0

```
A ? B ? C : D : E ;
```



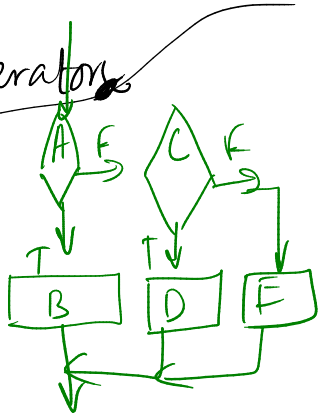
Operands ? `z = x + y;`

operators ?

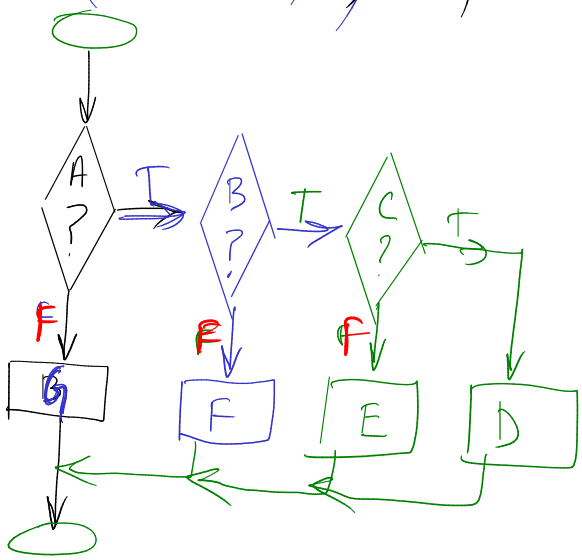
? : conditional operator

R → L

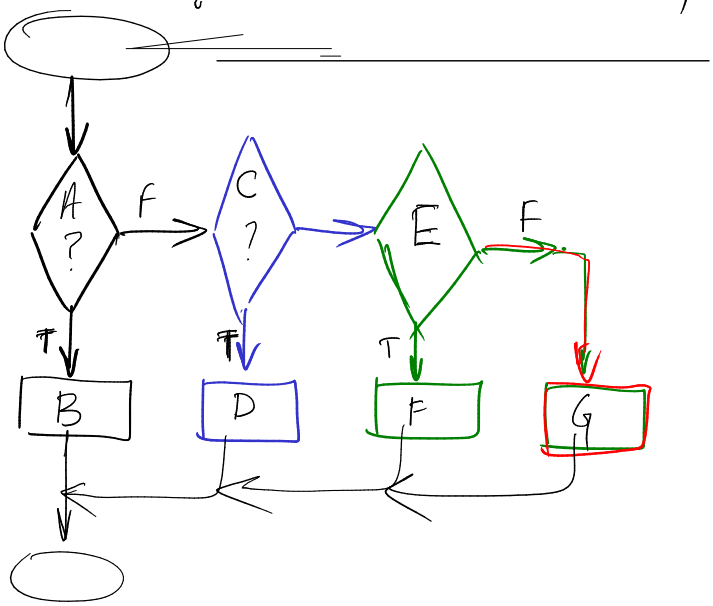
```
A ? B : C ? D : E ;
```



A ? B ? C ? D : E ; F ; G ;



A ? B : C ? D : E ? F : G ;



pf("%d", x&0); //0

, x&-1); //x

, x&1); 0 / 1

-1 => 1111 1111 1111 1111

& x => bbbb bbbb bbbb bbbb

x => 0000 0000 0000 000b

1 0000 0001 (1<<0)

& & &

x bbbb bbbb bbbb bbbb

0000 0000 0000 000b

result = x&1;

→ 0 : LSB of x is zero ; x is an even number

→ 1 : LSB of x is 1 ! x is odd Number

pf("%d", x&32); 32 / 0

→ Bit-5 in x is set

→ Bit-5 in x is clear.

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

32 0000 0000 0000 0000 0000 0010 0000 (1<<5)

& &

x bbbb bbbb bbbb bbbb bbbb bbbb bbbb bbbb

0000 0000 0000 0000 0000 0000 0000 0000 => 32

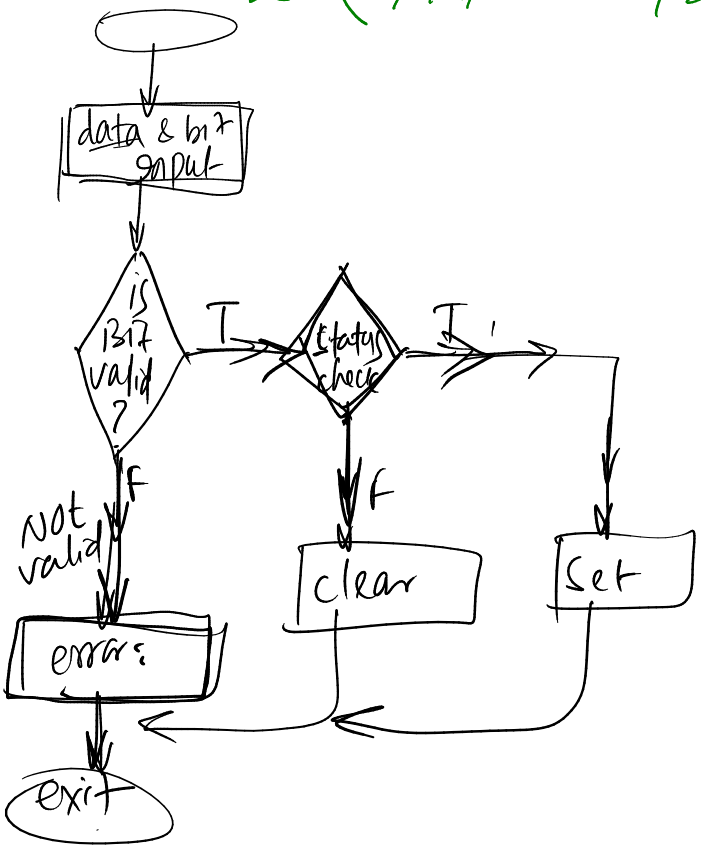
=> 0

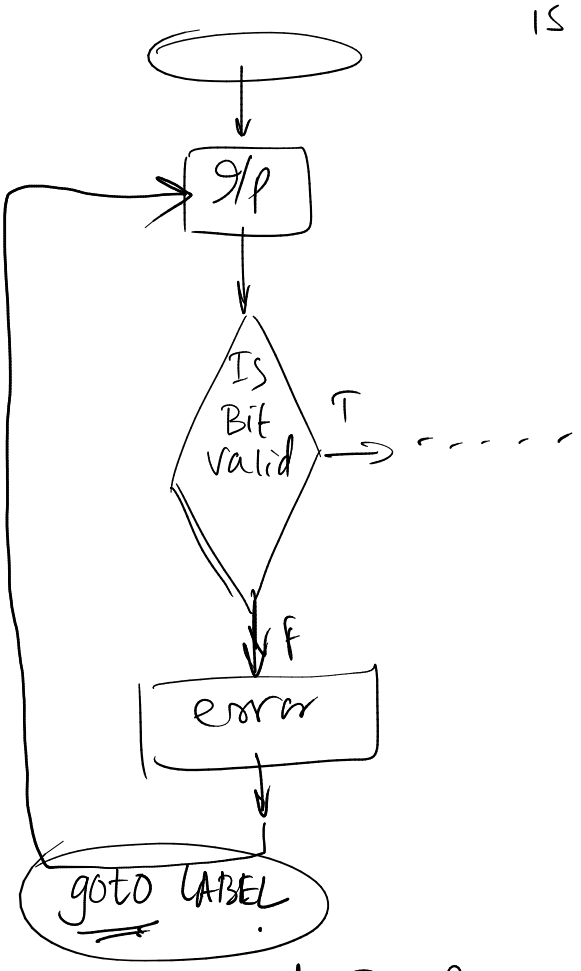
result = data &(1<<bit);

bit: 0 31

→ 0 ⇒ Zero: The bit in the data is clear.

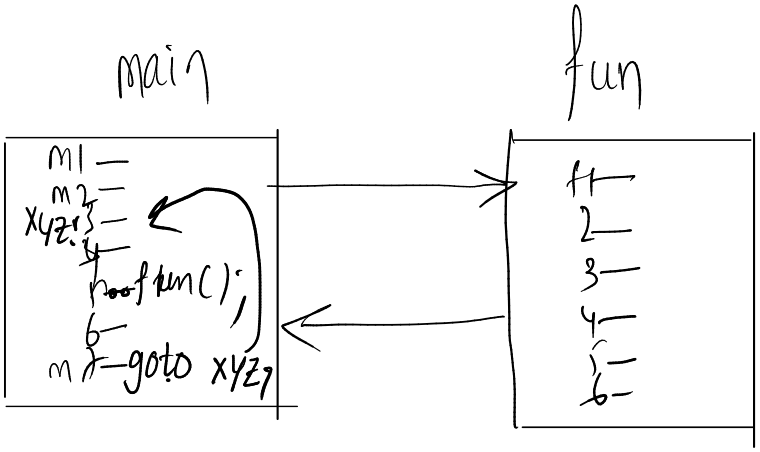
→ 2^{bit} (1/2/4/8 ..... $/2^{31}$) ⇒ Non-Zero: the bit in the data is Set.

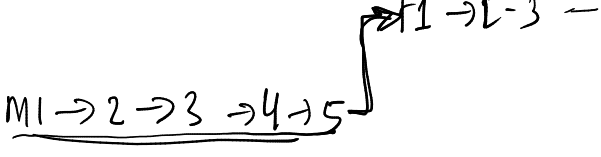




Internal Jump

Ext. Jump ⇒ function call.

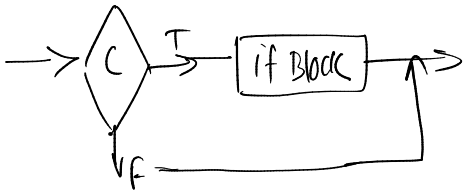




Branching with if

Syntax :

```
if (Boolean test condition)
{
=
=
=
=
}
```



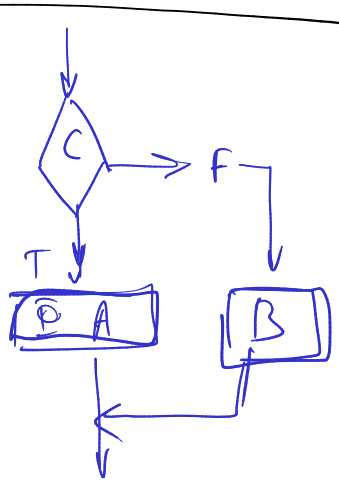
Branching with Conditional operator

→ Complex / Readability

→ Keywords NOT allowed

if ... else

```
if (condition) {
    =A
}
else {
    =B
}
```




```
if (bit is valid)
{
    if ( - )
        — set
    else
        — clear
}
else {
    error
}
```

Control stmt